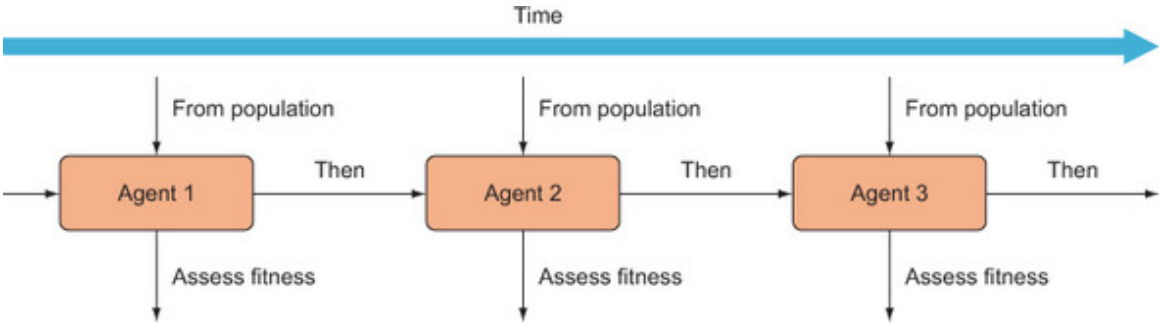
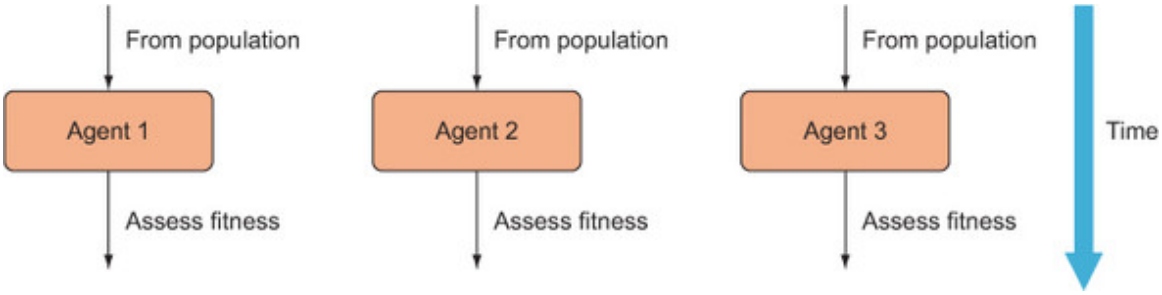


Determining each agent's fitness will generally be the longest-running task in an evolutionary algorithm, but each agent can evaluate its own fitness independent of each other. But there's no reason we need to wait for agent 1 to finish playing in the environment before we start evaluating agent 2. We could instead run each agent in the generation on multiple computers at the same time. Each of the 10 agents would go on 10 machines, and we can determine their fitness simultaneously. This means that completing one generation will take ~30 seconds on 10 machines as opposed to 5 minutes on one machine, a 10x speedup. This is known as running the process in *parallel* (figure 6.13).

Figure 6.13. If we have multiple machines at our disposal, we can determine the fitness of each agent on its own machine in parallel with each other. We do not have to wait for one agent to finish running through the environment before starting the next one. This will provide a huge speed up if we are training agents with a long episode length. You can see now that this algorithm is only a function of the time it takes to assess the fitness of a single agent, and not the number of agents we are assessing.



6.5.3. Scaling efficiency

Now we can throw more machines and money at the problem, and we won't have to wait nearly as long. In the previous hypothetical example where we added 10 machines and got a 10x speedup—a scaling efficient of 1.0. *Scaling*